

Self-Training with Structured Data for Efficient Insect Pest Detection

Bachelor's Thesis

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1 Introduction

Insect pests are responsible for significant production losses in agriculture [1]. However, pesticides threaten the environment by reducing biodiversity [2], particularly among insects, including pollinators that support agricultural production [3]. Overuse of pesticides causes resistance in pests. Additionally, global warming leads to increasing pest populations, which aggravates the problem further [1]. To address these challenges, the concept of integrated pest management (IPM) was developed. As defined by the Food and Agriculture Organization of the United Nations (FAO), it sets the goal of minimizing the use of pesticides by applying alternative management strategies wherever possible [4,5]. IPM is also the legal basis for the use of pesticides in the EU [6], which, among other things, prescribes the monitoring of insect pests when using pesticides. Precise monitoring facilitates the early and targeted use of pesticides for only small infested patches, the use of less harmful pesticides, or alternative ways of control like beneficial insects. Monitoring is usually based on setting up a grid of sticky traps, which are regularly checked for trapped insects. Because the manual counting requires expertise and takes a lot of time, it is not feasible for larger applications [7]. Hence, computer vision-based systems for automated monitoring of insect pests are considered to be part of the solution. Ongoing efforts to automatize insect pest monitoring have more recently adopted the methods of deep learning (see the surveys Teixeira et al. 2023 [8] and Mittal et al. 2024 [9]). My project aims to facilitate the implementation of deep learning-based, automated insect pest monitoring.

The dominant approach in pest monitoring in general involves the supervised training of YOLO models. YOLO ("You Only Look Once") is a deep learning architecture that unifies the tasks of object localization and classification into a single step. It was introduced by Redmon et al. in 2016 [10] and has been improved on since then in several iterations by different contributors. For greenhouses, specifically designed neural network models

have been employed successfully in 2021 [11,12], but the recent studies from 2023 onwards use YOLO models on individual datasets [13,14,15,16]. These six papers all report high performance, with mean average precision (mAP) or F1 scores ranging from 0.89 to 0.94.

Main challenges in the implementation are the small size of pests—which, even with high-resolution cameras, occupy only a very tiny fraction of an image—and the high degree of morphological variation among individuals. In less controlled environments additional complications arise from high morphological similarity between different species. These challenges create a demand for large annotated datasets; however, the collection and labeling of biological data require considerable effort and expertise. As a result, datasets are often small and exhibit other limitations, such as class imbalance. For an overview of these specific challenges, see Teixeira et al. (2023) [8]. As a remedy, the authors recommend semi-supervised learning (SSL), a family of methods for training with only partially data. Despite this, no survey reports applications of SSL in the field of pest monitoring in general and only one study in the context of the greenhouse makes use of SSL methods: Rustia et al. [17] proposed a self-training pipeline as a supplement to a fully supervised model as early as 2021. While the results were promising, the implementation was highly intricate, involving multi-stage setups with hand-tailored solutions. This complexity may explain why the approach has not been further explored. Outside of the greenhouse, in open-field settings, some efforts were made from 2024 onwards, either focusing on a single insect species with more diverse backgrounds or on a set of larger insects [18,19,20,21,22]. Another notable paper in this context is Dang et al. (2024) [23], who employ a field dataset of ten larger insect species and perform segmentation without manual annotation by leveraging a grounding language–image model. While extending this approach with classification and adapting it to the smaller insect pests typically found in greenhouses might be feasible, grounding models will not be considered in this study due to their limited speed and efficiency compared to YOLO-based methods.

Table 1 below shows an overview over the relevant literature. Green indicates a correspondence with the demands of my project, red the opposite, and orange overlappings. Because of the differing datasets and metrics, the results are not directly comparable and left out. Also not seen in the table are the pest classes included in the datasets: Only Rustia et. al [11] covered the same pest classes featured in this project. As of now no YOLO model has been trained to identify common greenhouse pests without relying on supervised training. Although working solutions for automatic insect pest monitoring in the greenhouse have been successfully tested, practical deployment stagnates and is hindered by the need for large, manually annotated datasets for new target domains. To my knowledge, there is no adequate method to reduce the amount of labels needed. The goal of the thesis is therefore an improvement in implementation efficiency by adapting the labeling and training method.

Authors (Year)	Greenhouse	YOLO	Training
Li et al. 2021 [12]	yes	no	supervised
Rustia et al. 2021 [11]	yes	no	supervised
Rustia et al. 2021 [17]	yes	no	SSL (predicted pseudo-labels by separate pipeline)
Costa et al. 2021 [13]	yes	yes	supervised
Zhang et al. 2023 [14]	yes	yes	supervised
Li et al. 2024 [18]	no	no	SSL (teacher-student)
Majewski et al. 2024 [19]	no	yes	SSL (pseudo-labels)
Wang et al. 2024 [15]	yes	yes	supervised
Zhou et al. 2024 [20]	no	no	SSL(teacher-student)
Gomez-Zamanillo et al. 2025 [22]	no	no	supervised, labels enhanced by image processing
Shi et al. 2025 [16]	yes	yes	supervised
Zhou and Shi 2025 [21]	yes	no	SSL(teacher-student)

Table 1: Overview over relevant papers. Colors indicate overlaps to my study (green: identical, orange: overlap, and red: different; yes/no indicate if the respective paper is settled in the greenhouse context or uses a YOLO-model respectively). None of these studies uses a simple self-training approach and none tries any type of SSL with a YOLO-model in the greenhouse setting.

I propose that given a specifically structured training dataset, a model can be trained without the need for manual labeling. This is achieved by a combination of image processing and a semi-supervised learning method (SSL) called self-training. The results are expected to contribute to a software foundation for a low-cost, open-source insect pest monitoring solution. It should be trainable on a single GPU and also provide efficient inference. Such a system could support practical deployment in greenhouse production settings and serve as a research tool for further studies.

The key requirement for the proposed solution is the specific structure of the dataset, which should be partitioned into subsets. All images in one subset should only contain one specific pest class as objects. With already existing labels a supervised YOLO model is trained as reference point, which should achieve at least 0.9 precision and 0.8 recall per class. Precision is prioritized in this target values over recall for two reasons. First, recall can be improved later through denser placement of sticky traps, whereas precision depends primarily on model quality. Second, low precision risks erroneous pesticide use, which conflicts with the principles of IPM.

The envisioned adapted automated labeling and SSL is done in two stages. First, with image processing segmentation algorithms bounding boxes for the pest classes are created. Since the images are presorted to contain only one pest class, the detected bounding boxes can be directly associated with the corresponding label, yielding a partially labeled dataset. In the second stage, the partially labeled dataset as given by stage one is used in a self-training SSL framework with an adapted YOLO to produce a model.

ToDo: Update this. At the moment the self-training does not work and I only created labels by image processing followed by one training with the created labels. So no SSL/self training really. But still without manual labels. Also adapt use case: It could be useful for implementing new insects to existing models.

2 Methods

2.1 Overview

:

- The main goal is to compare the SOTA-approach of supervised learning on manual labeled data with an alternative approach, that dispenses with manual labels completely.
- The alternative approach is separated into two main steps: (1) Creating labels with image processing and (2) Use this labels to train a model.
- The following steps were done for the comparison of the approaches:
- Data was inspected and prepared for experiments by cleaning and splitting into set, see Section 2.2.
- An evaluation pipeline was set up to test both models, see Section 2.3.
- Supervised training on manual labels was done as a baseline, see Section 2.4.
- Labels and augmented training data were generated by image processing, see 2.5.
- A model was trained on the generated labels (the model was used to predict new labels for more training iterations), see Section 2.6.
- Code and results can be found here: https://github.com/ChristophWald/insect_pest_detection, though not documented properly at the moment.

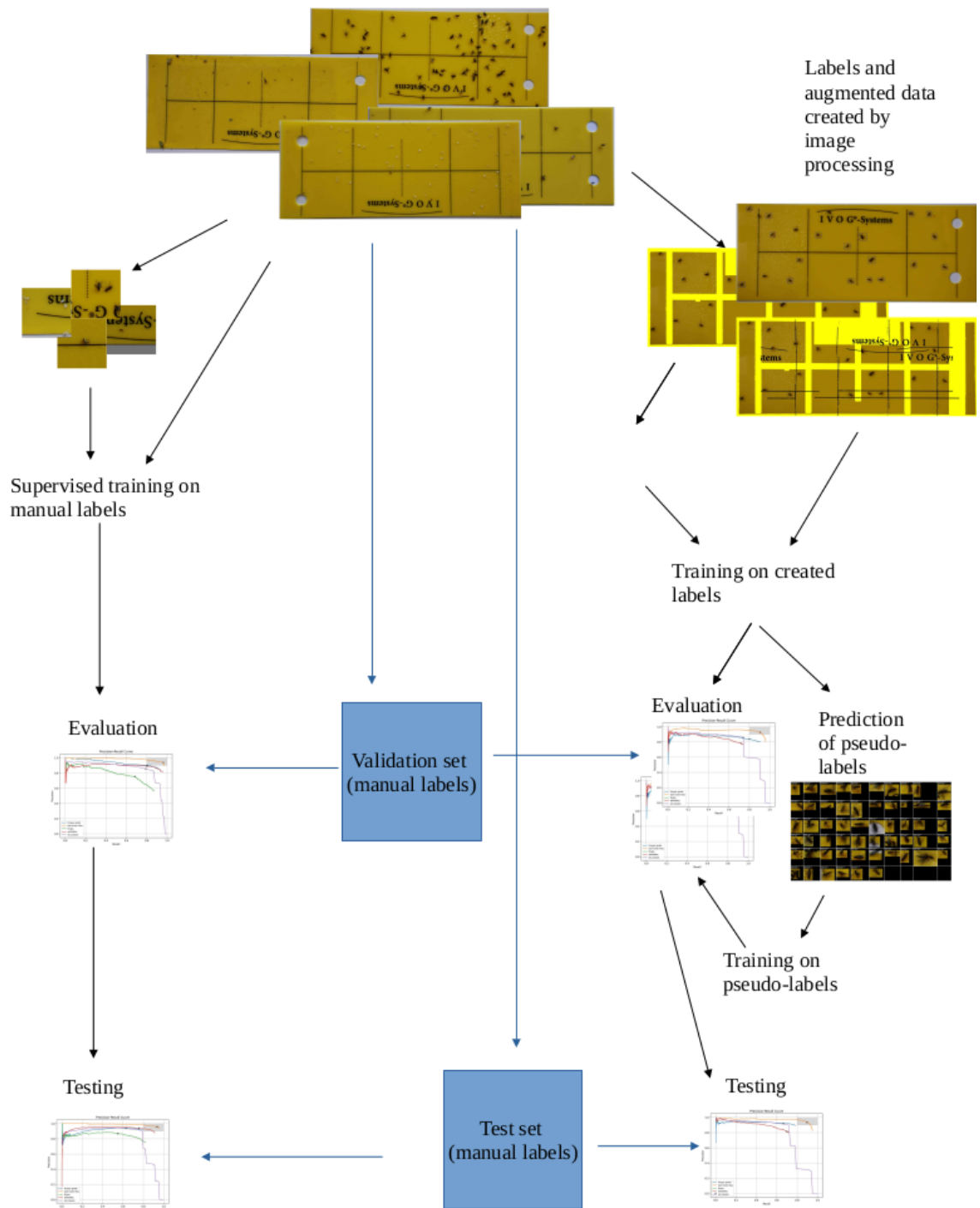


Figure 1: Overview over the experimental setup.

2.2 Data

2.2.1 Dataset description

The data is provided by the Julius-Kühn-Institut (JKI) and was collected as part of the Smart-Checkpots project [24] where it was labeled manually and used to train a supervised model. The images are divided into four subsets each associated with only one of four insect pest classes (see Table 2) to simplify the task of manual labeling. Every image in a subset is supposed to depict several instances of only one pest class trapped on a yellow sticky trap (YST)¹ As it is typical, the traps have a black background structure consisting of lines and an inscription, because the contrast lead to a higher attraction of insects.

Common Name	Scientific Name	EPPO Code
Fungus gnats	Bradysia impatiens, JOHANNSEN 1912 (Diptera: Sciaroidea)	BRAIIM
Tomato leaf miner	Liriomyza bryoniae, KALTENBACH 1858 (Diptera: Agromyzidae)	LIRIBO
Western flower thrips	Frankliniella occidentalis, PERGANDE 1895 (Thysanoptera: Thripidae)	FRANOC
Greenhouse whitefly	Trialeurodes vaporariorum, WESTWOOD 1856 (Hemiptera: Aleyrodidae)	TRIAVA

Table 2: Overview of insect pest classes present in the dataset.

The chosen insect pests are common in greenhouses, particularly on crops such as tomatoes and cucumbers, where they can cause severe damage, up to complete crop loss [11]. As Figure 2 shows, the size of the pest classes and thereby the number of pixels they cover differs considerable. In thrips, also male and female differ in appearance. The female individuals are larger and have good contrast against the yellow background, whereas the smaller and more transparent males are harder to detect (Row 3, box 3,7, and 9).

TODO: Change Row3/Image1

¹In general, there are different types of traps e.g. pheromone traps and there are also sticky traps with different colours, but YSTs are the most commonly used traps in greenhouses. Yellow is a compromise which attracts most insect pests.

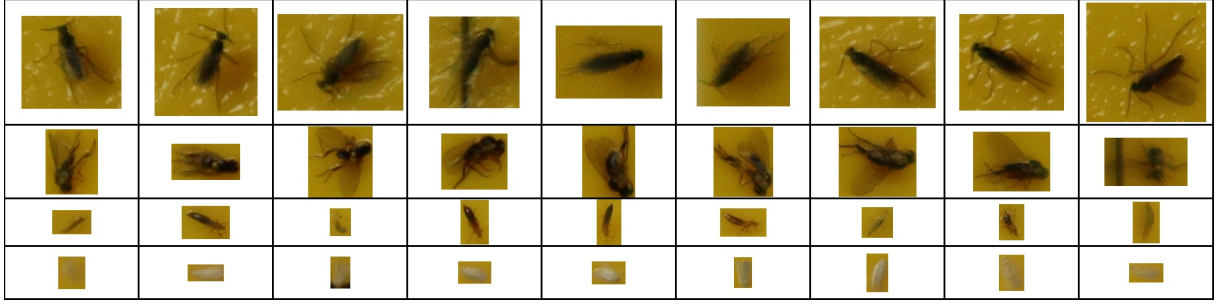


Figure 2: Cropped boxes with fungus gnats, leaf miner flies, thrips, and whiteflies (top to bottom). Original resolution.

The populated YSTs in this dataset were created at the JKI by breeding the insects to the adult stage in a closed environment (see Figure 3a). For the fungus gnats and the leaf miner flies the the YSTs were exposed in the environment for some time until it became populated. For the whiteflies and the thrips the tapping method was used, where the YST as tapped against the plant to catch the insects. The traps were then immediately attached onto a piece of white paper, and pictures were taken with a Canon EOS M50 Mark II on a tripod at a distance of approximately 13 cm (see Figure 3b) and a resolution of 6000×3368 pixels, resulting in files of about 5–7 MB. No artificial light was used to avoid reflections. The pictures have slightly different view angles and different lighting conditions, some still affected from reflections and some having areas with low sharpness. Some of these aspects may be also be part of the real data to be processed after training.



(a) Breeding station



(b) Camera setup

Figure 3: Data creation, pictures with kind permission M. Pollmann, JKI

While the YSTs were captured, few insects could leave the trap and were then trapped by another trap. Therefore the subsets can contain some insects belonging to other

subclasses. The tapping method for the whitefly also caught some parts of insects, but no foliage, because the tobacco plants are robust enough. Thrips however were bred on bean plants which left considerable amount of foliage on the YSTs. Also, thrips in larval stages are trapped, which are hard to distinguish from male individuals. In the use case, larvae will cannot be trapped, because only adult can fly.

For each pest class a proportion of images were manually labeled by several persons. Visual inspection showed inconsistency in the correctness of labels, the completeness of labels and the size of the bounding boxes.

2.2.2 Preprocessing

To raise the quality of the dataset, images and labels were checked and improved were possible.

Image files were sorted by their creation time given by the metadata and then the first 100 images for each class were visually inspected, which contained several images with identical YSTs and different camera setting as leftovers of early experiments with the camera setup. These were removed manually. I removed further duplicates (identical images) by comparing file checksums. Images with labels that indicated no insects or unexpected insects not belonging to the class of the subset were also removed.

Because of the reported inconsistencies in the labeling, the test set comprising 289 images was checked and corrected by an employee of the JKI, which took about x hours. In total x labels were added. Unfortunately time constraints prevented a relabeling of the training set, such that supervised training and all validations had to be carried out on a label set of lower quality. The results of the supervised training therefore provides only a lower bound for the expected quality of a model.

For the thrips, I inspected the complete subset. This led to the exclusion of 200 images with wrong labels or too much other objects. To improve the quality of the thrips labels further I gave the cut out bounding boxes to an expert from the JKI, which identified 1276 labels as incorrect. These were removed.

Table 3 shows the number of images and labels after all cleaning reported in this section. The dataset is imbalanced, both in terms of the number of labeled images per class and the mean number of instances per image.

Because of the summed difficulties the thrips images and labels posed, the class had to be discarded for all experiments but the supervised training, which showed that no satisfying

Class	Images	Labeled	Fraction labeled	Instances labeled	Mean instances per image
FungusGnats	1328	1018	76.66 %	33373	32.78
LeafMinerFlies	1422	659	46.34 %	6518	9.89
Thrips	1134	392	34.57 %	5453	13.91
WhiteFlies	1105	433	39.19 %	25735	59.43
Total	4989	2502	50.15 %	71527	28.59

Table 3: Annotation statistics after cleaning and preprocessing.

results are to be expected in the proposed self-training. There are three major reasons for this decision: low sample size, presence of non-target objects, and low label quality. Because of size and morphological differences between adult males and females the thrips is the hardest pest class to detect and has to be represented with an according number of images and instances in the dataset. Unfortunately, after cleaning the thrips subset has the lowest number of images and labels in the dataset. Furthermore the thrips are easiest to be mixed up with other objects like foliage and therefore require clean images. But the thrips images are the only pest class that has lots of non-target objects, including thrips larvae on their images. The 1276 false labels clearly show that the automated labeling cannot work under the assumption that all objects on an image are instances of the respective pest classes. Finally, the inconsistent labeling quality prohibits proper testing.

Some labels were found to contain negative coordinates. After visual inspection of samples, these values were interpreted as their absolute counterparts, since this matched the actual object positions. The underlying cause of the negative coordinates could not be identified but seems to be connected to the export of labels from Label Studio.

2.2.3 Splitting

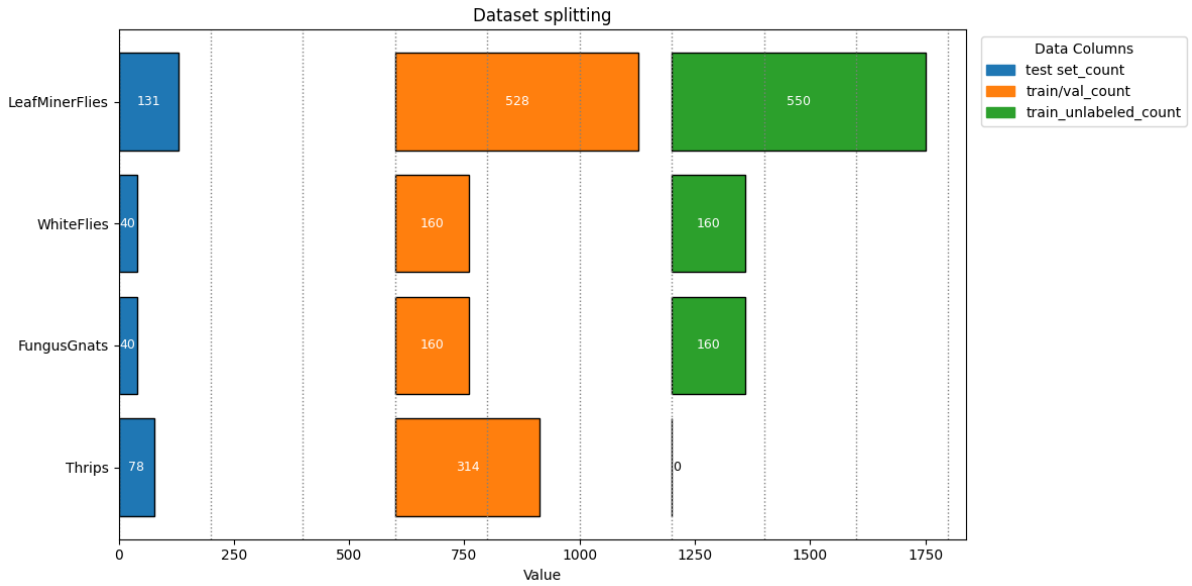


Figure 4: Number of images in the randomly created subsets. Sets are balanced because the absolute number of species per images is different for each class.

Figure 5: Data splitting

To balance the labeled sets, the labeled images for the fungus gnats and the white flies were reduced to 200 each, because these pest classes have considerably more instances on a single image than thrips and leaf miner flies (see 3. The other labels are discarded and the images retained as unlabeled. The labeled set is then further divided randomly into 20% test set and 80% training set. From the unlabeled images excluding the thrips a set is drawn with comparable sizes as the training set.

2.3 Evaluation

For the evaluation the predicted labels and bounding boxes are first filtered for overlapping boxes. The remaining predictions were sorted by confidence threshold to produce the precision-recall curve produced. True positives are given by intersection of union (

$$\text{IoU} = \frac{|A \cap B|}{|A \cup B|}$$

) of bounding box and ground truth box, but also by boxes being contained in the ground truth box. The point on the precision-recall curves being closest to the target zone of precision ≥ 0.9 and recall ≥ 0.8 was taken as operating point for the class. Different

models were then compared by the number of points in the target zone, or if equal, by the remaining distances to the target zones, or if not given or equal, by the best F1 value. For the tiles the prediction was done with a window sliding over the image.

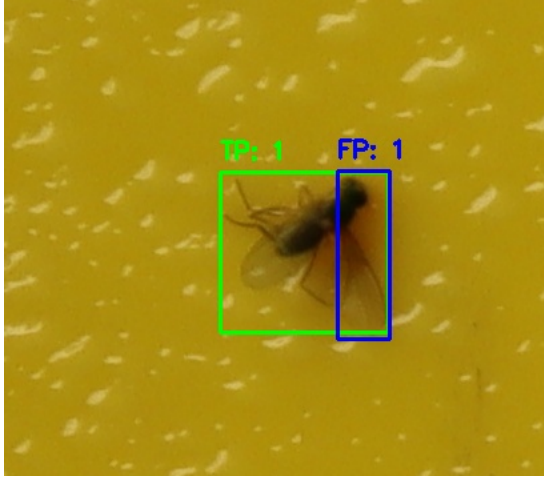
This section should contain:

- the sliding window prediction + nms + containment but without confidence threshold to get all predictions
- testing of stride 420/440 and containment filter on/off (see results, for stride include images from `cpu_vs_gpu.ipynb`)
- comparing the boxes to the ground truth with thresholds for iou and another containment filter, greedy matching to allow only one true positive per GT box
- testing containment off/0.9/0.5 (see results)
- building of the prec/recall curve, find best thresholds by: a) if prec/recall arrive target values: take best f1 or b) take prec/recall with lowest distance
- sliding window prediction with per class confidence threshold applied to predict boxes
- for the evaluation of the image processing only the comparing to GT is needed
- best model found by comparing a) number of classes that reached target metrics, b) distance to target metrics area, c) f1 values
- filters build on visual inspections of the false positives/negatives

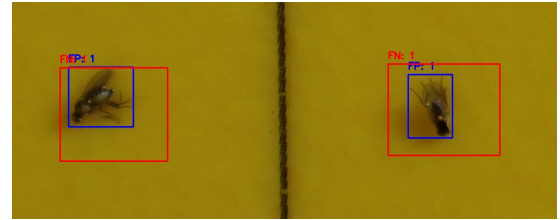
Non-maximum suppression (NMS) is used to filter overlapping bounding boxes for the same object. It relies on the intersection of union (IoU) metric, which is the ratio between the area where two boxes overlap and the area that the boxes cover when combined. When two boxes have an IoU above a threshold, only the box with the higher confidence is kept. However, if a small box is completely or partially inside a larger box, the IoU will be low, because the intersection is only as small as the small box and the union is as big as the large box. As visual inspection shows, the small objects in the dataset make it necessary to include an extra filtering for this case. IoU for NMS was set to 0.4 and the threshold to filter out the other case was set to 0.5. See Figure 6 a for false positives that occur without the extra filter.

Comparison to the ground truth also relies on IoU and as usual, predicted bounding boxes were counted as true positives with an IoU equal or larger than 0.5. Again, visual

inspection shows small correct predictions inside larger ground truth boxes are not covered by this procedure. In an additional routine, also predicted boxes contained at least with 0.5 of their area in ground truth boxes are counted as true positive. See Figure 6 b for false positives that occur without the extra comparison.



(a) Overlapping bounding boxes not detected by NMS (leaf miner fly), which creates a false positive box.



(b) Bounding boxes counted as FP because of low IoU with the larger ground truth box (leaf miner flies).

Figure 6: Detection errors caused by low Intersection over Union (IoU) with small boxes. Green boxes are true positives (TP), blue boxes false positives (FP), red boxes false negatives (FN).

2.4 Supervised training

I used a YOLOv8 model. I trained on the full images resized to 1024 internally by YOLO and by two different sets of 640x640 segments built with stride 440. The sets differ by the inclusion strategy: The first set simply includes all resulting tiles. In the second tiling approach, for each object only one tile was allowed to avoid redundancy in the training set and tiles with partial objects were dropped. Additionally for the first set the proportion of empty background images included and the exclusion of the thrips were tested. Augmentation was applied only carefully.

As a reference point for the automated labeling process, a YOLOv8 model is trained using the manual labels provided. The aim is to demonstrate that the data is sufficient to reach the target metrics of a minimum per-class precision of 0.9 and recall of 0.8. Since supervised training is not the main focus of this project, experiments are not designed to fully optimize results. A small YOLOv8s model pretrained on COCO is chosen because initial tests showed that larger models or later versions did not improve performance. With 4-5 GB of weights, the small model also meets the requirement of single-GPU training and

the more modular code of version 8 is easier to adapt for self-training. No task-specific pretrained models are available to my knowledge.

The images are cropped to the size of the YST with color threshold based segmentation (compare YSTs in Figure ?? and Figure ??). This decreases the size of the dataset by 17,1 GB but also introduces different image sizes. Labels are therefore recalculated after cropping. Since YOLO labels are relative to image width and height, the new image dimensions must be used when transforming labels back into absolute coordinates.

YOLO expects images of size 640x640 and resizes images with larger resolution. Because the insect pest classes are of small size, this resizing might cause difficulties. Therefore images are tiled. After padding each image such that its dimensions are divisible by 640, the images are tiled into 640×640 patches with a stride of 440. The overlap of 200 pixels corresponds to the size of the largest pest class, the fungus gnats, such that an instance is at least contained in one tile. The labels are recalculated according to the tiles and labels and are only kept if they are contained to at least 80% inside the respective tile. The resulting tiles contain a large number of empty background tiles, which account for over 68% of the tiles in the labeled training set. For the training on the full images, the rescaling parameter in YOLO was set to 1280.

Starting from the default settings, short training runs with selected parameter changes were conducted to identify optimization potential, where augmentation settings and the ratio of background to foreground images are identified as the main factors influencing performance. Changing the default optimizer (stochastic gradient descent), learning rate schedule, and loss weights did not yield improvements. Because the dataset already contains considerable variance (see Section 2.2.1), augmentations are applied only subtly. The scaling factor is reduced to avoid unrealistic size differences between objects, and the probability of mosaic augmentation is lowered to prevent the model from learning lighting differences between images. A small amount of mixup is introduced with a 5% probability. All other parameters are kept at the default values. (I could put the full list of arguments in the appendix).

All models were trained up to 200 epochs, then the best epoch (in terms of mAP@90:95) was taken.

Training run	Dataset	Background ratio	epochs trained	Best epoch
train1	tile v1	68%	200	144
train2	tile v1	30%	200	127
train3	tile v1 without thrips	200	47%	136
train4	full images	0 %	100	94

Table 4: Selected training runs for supervised training

For the training runs the proportion of background tiles (tiles without objects) is varied (see Table 4). The initial training contains 68% background tiles, which is reduced to 10% and 30% in a number of training runs. To test the effect of the thrips images on the training of the other classes, the set with only 30% background images is stripped from all images from the thrips subset which changed the background to foreground ratio to 47%.

2.5 Image processing

The aim of the image processing is to detect a proportion of objects and create labels including bounding box coordinates, which will serve as the starting point for semi-supervised training of a YOLO model. The simple main idea of this step is to separate background and foreground with a color threshold and then keep all coherent foreground contours as objects. A rectangle around the contour serves as bounding box and the class id can simply be derived from the subset of the image, as all insects on an image are of the same pest class. For images without background structures the process reduces to the short and fast procedure reported in section 2.5.2. However, when images contain background structures—as in the dataset used in this study—more complex preprocessing is required, as detailed in Section 2.5.1. The presented pipeline was developed with the labeled training set. In between evaluation is done on the ground truth given by the manual labels. Raising the absolute number of true positives and minimizing the number of false positives are the main metrics. Precision is also tracked but is considered secondary, as it might come at the cost of too less true positives.

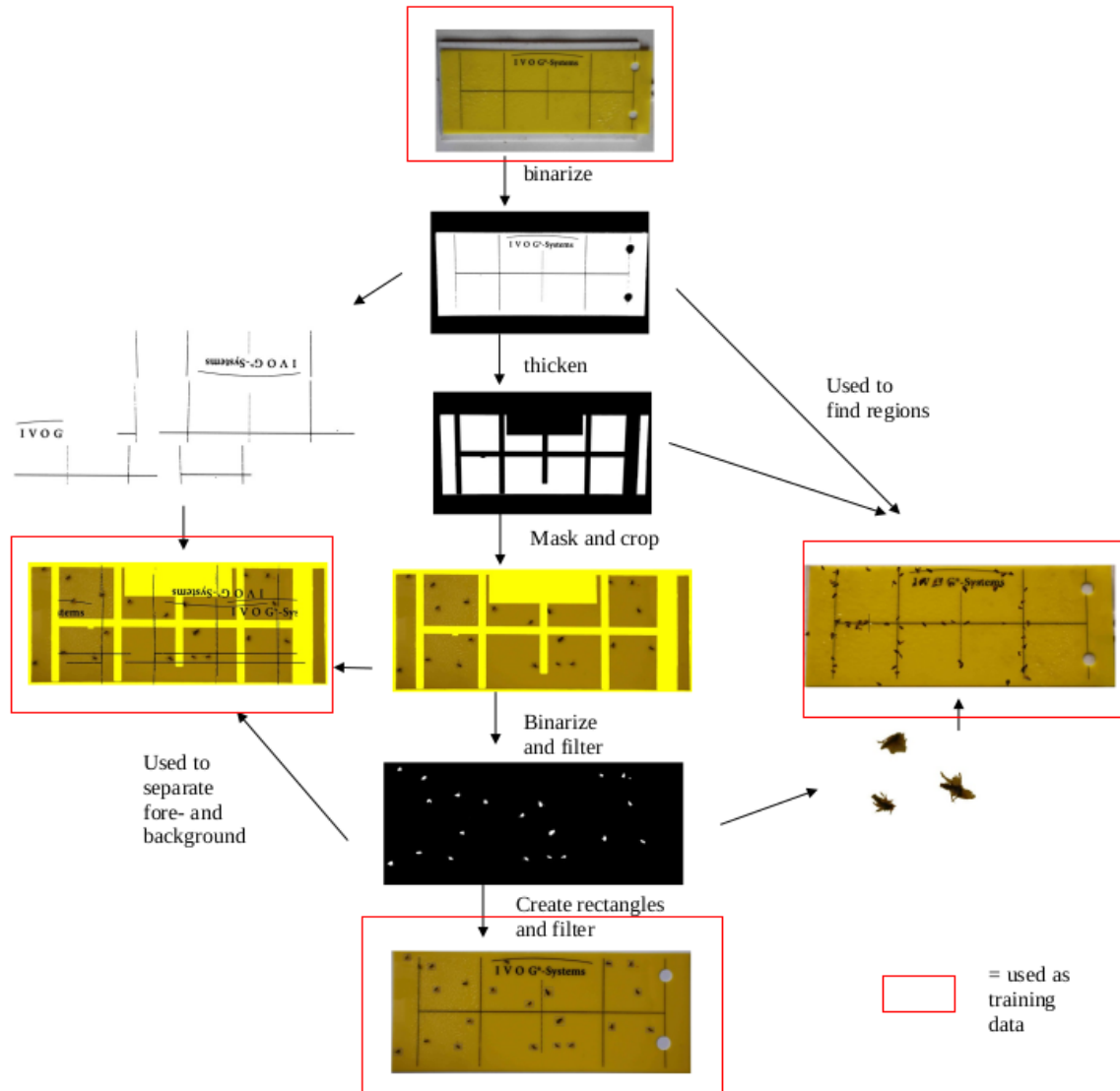


Figure 7: Overview over creating training data with image processing.

Possible example images not included: LIRIBO_0614 for effect of shifting the amsk, TRIAVA_0037 for effect of fatter maks, TRIAVA 0320 for missing unsaturated whiteflies, TRIAVA 506 for the detrimental effects of other filtering, all files in image Processing evaluation - mask tests

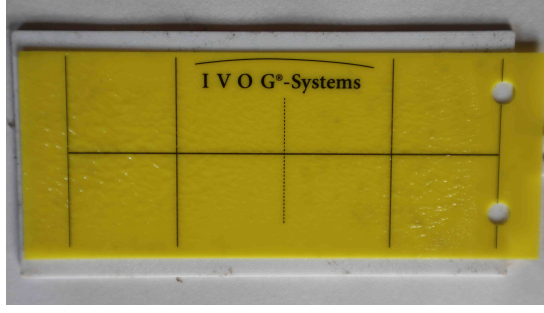
2.5.1 Preprocessing for segmentation

I created some masks (Figure 8) and devised a processing pipeline. The result is a binarized image which divides insects as foreground and the black structures as background

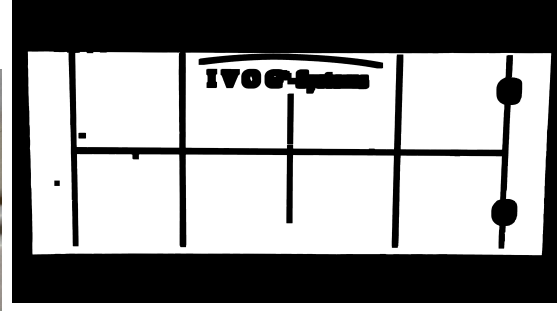
To filter out background structures, a mask covering these regions is applied to each image. The pixels within the mask are changed to a yellow hue, allowing them to be identified as background during segmentation. Direct detection of the lines was tried, but led to no useable results, as the detected line segments were often very short and could not be separated from the line segments of the objects, that should be kept. Because the images in the dataset were captured from slightly different angles, the mask must be aligned individually for each image. All image processing was performed using the cv2 Python library. Binarization, which is used at several steps, is achieved through color thresholding in the HSV color space, as the hue component provides robustness against variations in lighting. The thresholds used were: Hue 20–30, Saturation 100–255, and Value 100–255.

The processing pipeline begins with a subset of uncropped images, as determined by the splitting described in Section 2.2.3. Since the masking procedure assumes the header is located at the top of the YST, images were rotated when necessary. The required rotation was detected by dividing the YST area into horizontal halves and comparing the summed black pixel area in each half, with black pixels defined as having a value below 50. This method becomes unreliable when the YST is crowded with a large number of black insects, so a visual inspection was performed to ensure accuracy. Correspondingly, the labels used for manual evaluation were rotated to match the corrected image orientation.

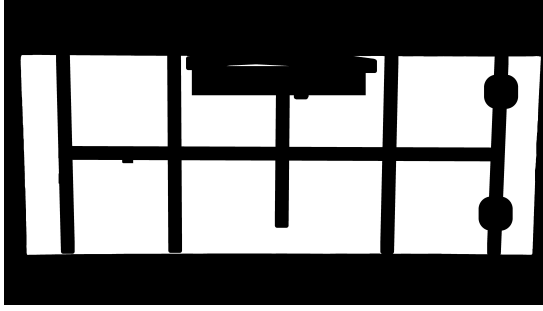
Creating the mask required balancing line thickness to fully cover background structures despite minor misalignments, while avoiding excessive coverage of the objects. Binarized and denoised images of empty YSTs, captured under the same conditions as the dataset images, were used for mask generation. The masks were created by combining several empty YST images with projective transformations and thickening the black structures with dilation (kernel size 25) (Figure 8b). From this mask, two additional variations were generated by applying further dilation (kernel size 50) (Figure 8c) and by manually thickening the right border lines to cover the holes completely and drawing a rectangle over the header (Figure 8d). For subsequent alignment, the corners of the YST contour and two points of the horizontal midline calculated from the original masks were saved.



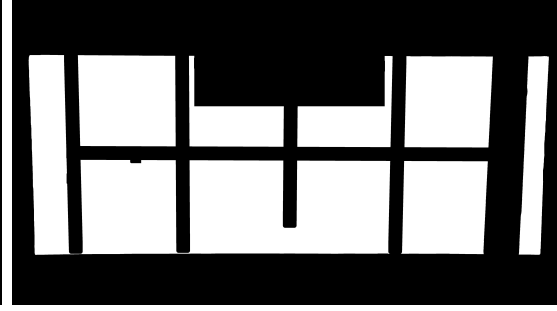
(a) Image used for the creation of masks.



(b) Mask created by overlaying other images to the initial one and then thickening the lines.



(c) Mask from b) with thicker lines.



(d) Mask from e) with a rectangle as header.

Figure 8: Masks tested for masking out the background structures.

With the masks prepared, preprocessed and masked versions of the image set were created. The pipeline of this process is divided into the following steps.

- Find contours on the binarized image. Take the largest, which is the YST. (algorithm used: `cv2.findContours`)
- Assuming a rectangle shape find the corners or exclude the image from further processing, algorithms used: `cv2.arcLength`, `cv2.approxPolyDP`)
- Using the corners of the mask and the image, do a projective transformation of the mask and of the midline of the mask. , algorithms used: `cv2.findHomography`, `cv2.warpPerspective`)
- Using the middlepoints of middle horizontal lines of the mask and the image, do another affine transformation of the mask, if the difference is lower than 500 pixels. If the result is larger, a wrong horizontal line was picked and the transformation is skipped. , algorithms used: `cv2.HoughLines`, `cv2.perspectiveTransform`, `cv2.warpAffine`)
- Color the pixels given by the aligned mask in the image yellow (background) as seen in image .

- Crop the image to the YST (and recalculate the manual labels for evaluation).

2.5.2 Segmentation

The foreground objects need to be filtered further. For testing the masks this was done by simply thresholding the size of the foreground contours. For final use a more refined set of thresholds is derived from the distribution of the sizes and shapes of the rectangles created around the contours. (see Table 5).

More intricate binarization for TRIAVA was tried, but was not successful. Value thresholding for TRIAVA was tried but was not successful.

The segmentation is done on the binarized, preprocessed images. The results of the segmentation are filtered by several thresholds. A first set of threshold is used to evaluate the effects of the different masks as seen in Figure 8. A second, more refined set of thresholds is derived from inspecting the data given by the best evaluated mask (Figure 8d. Evaluating was done on the manual labels of the labeled training set. Comparing to GT was reduced to IoP, because it gives nearly identical results for smaller predictions inside ground truth and is more strict for overlapping ones.)

The first set entails guessed lower and upper threshold for the area contours detected. For fungus gnats and leaf miner flies only contours with an area in the range between 1000 and 10000 pixels are kept, for the whiteflies the range is between 100 and 1000 pixels. The rectangles around the contours kept are then scale with factor 1.5, because the contours lost small features like legs, that would cause the rectangles to be too small (see Figure ??). A second threshold then filters the rectangles by the ratio of their height and width. Assuming that more oblong rectangles are likely to be non-insect objects (e.g. left over background elements, foliage, parts of insects), the maximum ratio was set to 2. This excludes rectangles with one side being more than twice as long as the other side.

For each mask, segmentation is done for all preprocessed images and the results were evaluated by comparing to the manual ground truth with the Intersection over Prediction metric with threshold 0.5. The results from the best performing mask Figure 8d are used to inspect the distribution of box sizes, box ratios and, only for the whiteflies, the value in the HSV-space for all rectangles. Additional filter thresholds were then tested on the calculated rectangles that are given in Table 5. Goals were to reduce the number of small insect parts and occasional non-insect objects, as well as the number of large boxes with more than one insect occluding each other. Only for the whiteflies a considerable number of overlapping boxes were created, presumable because individual objects often led to two separate contours. Therefore this case was checked and only one rectangle kept.

Species	Threshold	Filter set 1	Filter set 2
Fungus gnats	Min contour	1000	<i>same</i>
	Max contour	10000	<i>same</i>
	Max box size	-	95th percentile
	Max box ratio	2	95th percentile
Leaf miner flies	Min contour	1000	<i>same</i>
	Max contour	10000	<i>same</i>
	Max box size	-	95th percentile
	Max box ratio	2	95th percentile
Whiteflies	Min contour	100	<i>same</i>
	Max contour	1000	2000
	Max box size	-	95th percentile
	Max box ratio	2	95th percentile

Table 5: Thresholds of the two filter sets for filtering the results of the segmentation.

2.5.3 Augmenting the data

Masking the background structures including the insects on it will create biased label set and the model learns to ignore insects on background structures. To prevent that, two types of augmentation sets with insects and lines overlayed are created. In the first set the masked image set is overlayed randomly by background structures. In a second set insects detected by image processing are cut out and placed on the background structures on empty YSTs (see 7 for examples).

Receipt for augmentation set 1

- take the masked images and divide into background and foreground by segmentation
- take an empty YST and get the background structures segmentation
- cut the borders to not have the black margin and the holes, make a random shift such that the structures are placed on a random location on the image
- does this a second time but with a rotated empty YST
- foreground original image -> background structures -> background original image to have the insect on top of the lines
- was done for the complete training set to replace the original images

Receipt for augmentation set 2

- take empty YSTs and shear them to have more variation
- take the bounding boxes detected by image processing and make a more refined background/foreground segmentation (for whiteflies I do segmentation by fixed threshold, then i find the white pixels inside the contour and restore them to the original color, for fungus gnats I tried fixed threshold segmentation and segmentation with a value threshold derived by otsu's method (v1/v2); I also filtered the remaining contour to keep only the largest)
- for each YST: find the background locations by masking, choose some random insects (per class) and place them there (i used a mask for the lines and another mask for the area around the mask to cover both cases; i also took care, that insects are on all relevant structures - not the holes - by dividing the image into segments and distributing them there). For the whiteflies I only introduced value variations in ranges taken from inspecting the true positives and false negatives from image segmentation.
- additional augmentation to variate the YSTs: lighting and orientation
- for each pest class 24 images were created to supplement the training data

2.6 Self-Training

I first trained on the image processed labels with augmentations, where the model learned to detect objects and classify them. I then used this model to predict labels on the original dataset without augmentations such that also insects on the background structures are now labeled. I then tried to improve the model by adding more pseudo labels.

2.6.1 Training on the labels created by image processing

Run	Flipud	Epochs	Basic Data	Empty Tiles	Augmented Set 2
train1	no	10	tiles	no	no
train2	no	10	tiles	yes	no
train3	yes	20	tiles	yes	no
train4	yes	20	tiles	yes	v1, on lines
train5	yes	20	tiles	yes	v1, on and near lines
train6	yes	20	tiles	yes	v2, on and near lines
train9	yes	20	augmented tiles 1	yes	no
train10	yes	20	augmented tiles 1	yes	v2, on and near lines
train14	yes	20	augmented 1 full images	no	full images v2, on and near lines

Table 6: Training runs for training on the image processed labels. Flipud turns the image 180° with probability 50%. Tiles is the unaugmented dataset with all tiles containing created labels. Augmented tiles 1 is the dataset as described in... Augmented set 2 are the augmentation described in Rescaling for full images was set 1280 as in supervised training.

- was done on tiles (as explicated in the section on supervised training)
- Learning rate (0.00125) and Optimizer (AdamW) were set automatically by YOLO.
- only tiles including at least one labeled instance were included in the basic train set
- for validation subset the manual labels are kept
- augmentation setting of yolo were taken from above (filter set 2), but also rotation was tested
- tests on inclusion of empty tiles
- tests on replacing the original dataset with the augmented set 1
- addition of augmented set 2 tested
- should be compared to full image training
- should be compared to the results on the unlabeled set

2.6.2 Further training on predicted labels

Run	thresh fungus gnats	thresh leaf miner flies	thresh whiteflies
train11 0.25	0.25	0.25	
train12 0.8	0.8	0.7	
train13 0.7	0.7	0.6	

Table 7: Training runs for the self-training with additional pseudo-labels filtered by class-specific confidence thresholds for tiles. Flipud was always set to 0.5.

Run	thresh fungus gnats	thresh leaf miner flies	thresh whiteflies
train15 0.25	0.25	0.25	
train16 0.5	0.8	0.7	

Table 8: Training runs for the self-training with additional pseudo-labels filtered by class-specific confidence thresholds for on full images. Flipud was always set to 0.5, rescaling to 1280.

- same settings as before (learning rate, optimizer, epochs = 20, flipud 0.5)
- new labels were included as pseudo-labels filtered by a confidence threshold
- predicted class was hardcoded from the species given by the filename/datasubset on the first run, where the labels were transferred from the training on the image processed labels to the original dataset with backgrounds included
- epoch length 5 & 10 was tested
- inclusion of new images with new predictions was tested (on unprocessed images)
- weighing of the losses for the pseudo-labels (with disabled mosaic/mixup augmentations) was tested

3 Results

3.1 Comparing supervised training and self-training

The best overall model is the supervised model trained on the manual labels provided. In validation three pest classes arrive at the demanded precision and recall. Because

confidence thresholds for predicting are derived from the incompletely labeled training/validation set, it still misses the target zone for all but one pest class in the test set. The best self-trained model only arrives at the demanded metrics for one pest class and also shows weaker results in the validation.

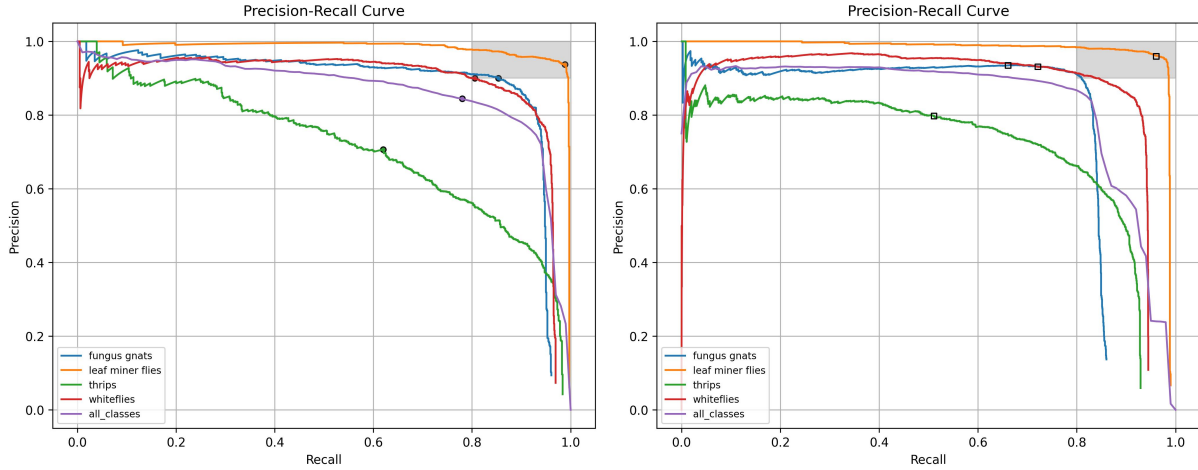


Figure 9: Precision-recall curves for the best model trained supervised (on full images), the grey rectangle defines the target zone. The left curve is taken on the validation set with the confidence thresholds closest to the target zone marked with circles. The right curve is taken on the test set with the operating points taken from the validation set marked as squares. Training the model on the incomplete labels would still provide a sufficient model for all but the thrips, however deriving the confidence thresholds from the incomplete labels makes the model miss its best performance on the completely labeled test set. Also the curves for all but the leaf-miners show higher precision on the test set possibly due to the more coherent labels.

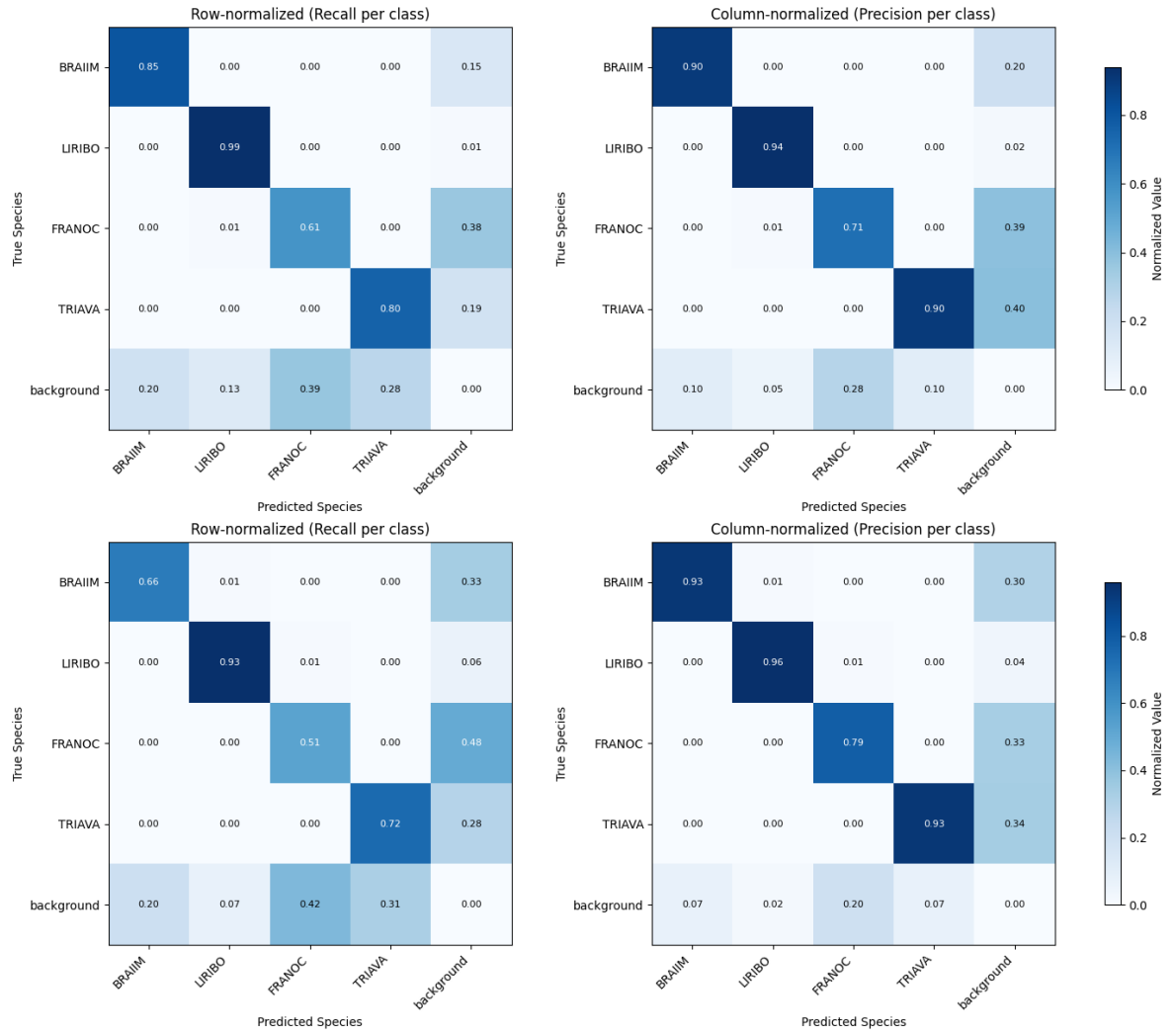


Figure 10: Confusion matrices for the supervised training on the validation set(upper) and the test set (lower). The thresholds derived from the validation set lead to higher precision and lower recalls in the test set.

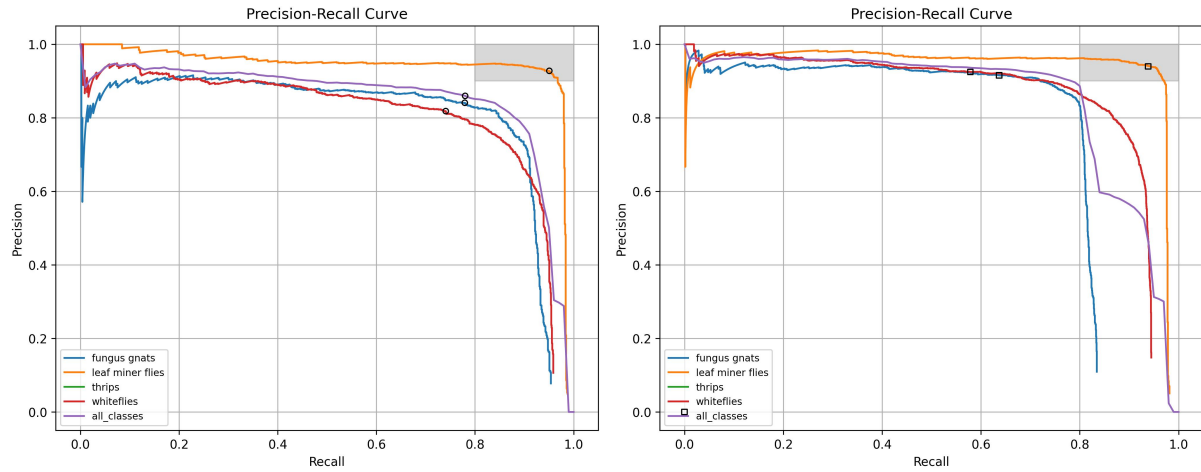


Figure 11: Precision-recall curves for the best model self-trained (on full images), the grey rectangle defines the target zone. The left curve is taken on the validation set with the operating points closest to the target zone marked as circles. The right curve is taken on the test set with the operating points taken from the validation set marked as squares. Self-training did not provide a sufficient model. (I could mark the points for the image processing to show the improvement).

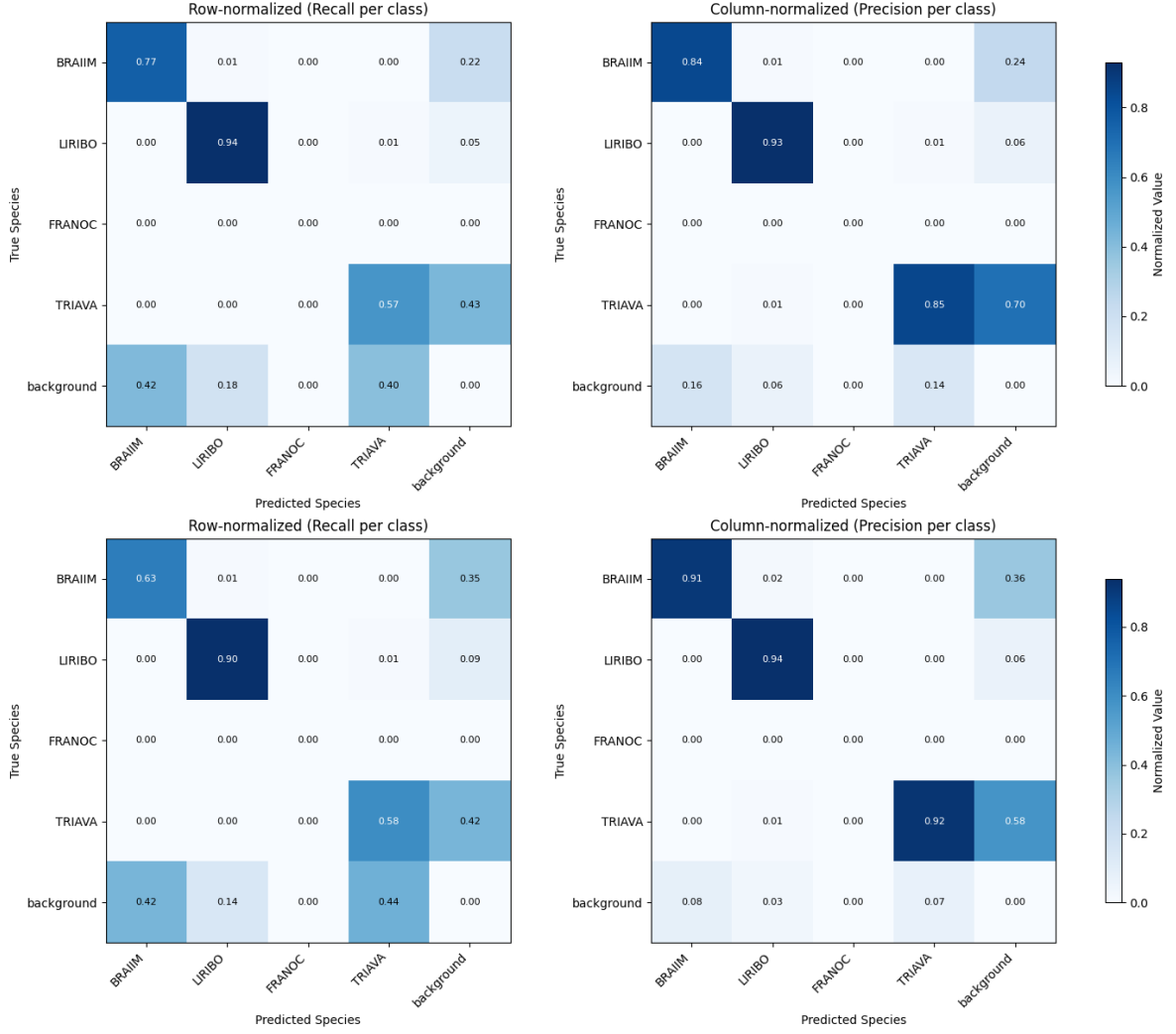


Figure 12: Confusion matrices for the self training on the validation set(upper) and the test set (lower). Again the thresholds derived from the validation set lead to higher precision and lower recalls in the test set. Both precision and recall are improved compared to the image processed labels (cf. Fig 16), with recall including now the insects on the background structures, that were masked out before.

3.2 Evaluation settings

The additional filtering of the predicted boxes and the additional checking for contained boxes in the comparison to the ground truth allows better detection results.

Stride	Containment filter	Containment comparison	train1			train2		
			TP	FP	FN	TP	FP	FN
440	-	-	3240	637	1024	3469	664	798
440	0.5	-	3240	592	1027	3469	664	798
440	0.5	0.9	3331	501	936	3469	584	798
420	0.5	0.9	3569	540	698	3737	716	539
440	0.5	0.5	3601	508	666	3469	587	798
420	0.5	0.5	3601	508	666	3737	637	530

Table 9: Comparison of detection results under different settings of extra thresholds with given absolute numbers for true positives (TP), false positives (FP) and false negatives (FN). The models are trained on tiles, each 100 epochs, train1 on the full trained label set, train2 with only 30% of the empty background tiles. The chosen parameters in the last row improve results.

Confidence threshold is set to 0.5, non-maximum suppression threshold for filtering before the ground truth to 0.4 and IoU threshold for comparison to ground truth to 0.5 in all tests. Changing the stride to 420 always brings a boost for the number of true positives, the extra containment filter before comparing the boxes brings a slight reduction in false negatives only for train1. The extra containment section in the comparing to the ground truth with 0.9 always brings benefits in all three metrics, but setting the threshold to 0.5 only improves results for stride 420.

3.3 Supervised training

The supervised model on trained on the tiles was slightly worse then the one trained on the full images.

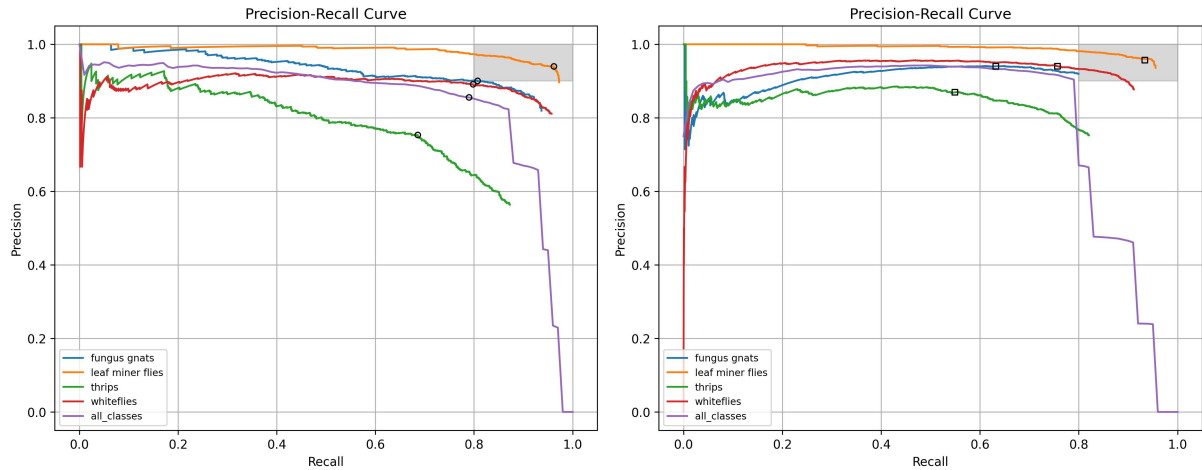


Figure 13: Precision-recall curves for supervised training on tiles. The results are lightly worse than supervised training on full images (see 9) for all but the thrips. Again the model suffers from the confidence thresholds derived from the insufficiently labeled validation set.

3.4 Image processing

The first subsection gives the results for testing the different masks that are needed to filter out background structures. In the second subsection the results for the different sets of thresholds that are needed to filter the rectangles created by the image processing are reported.

3.4.1 Preprocessing for segmentation

Topic sentence: The masks are used to filter out the background structures from the image before segmenting it to find the insects. The best mask decreased the number of false positives while retaining a sufficient number of true positives for the self-training.

Figures&Tables: Results for the different masks and the application procedures can be compared in Figure 14. Thresholds used for creating the results are found in the methods section, Table 5, filter set 1. Using this filter set and the mask with the best results, statistics were taken and used as threshold for filter set 2 (see Table 10).

Exact results in (mask_testing_collected_results.csv)

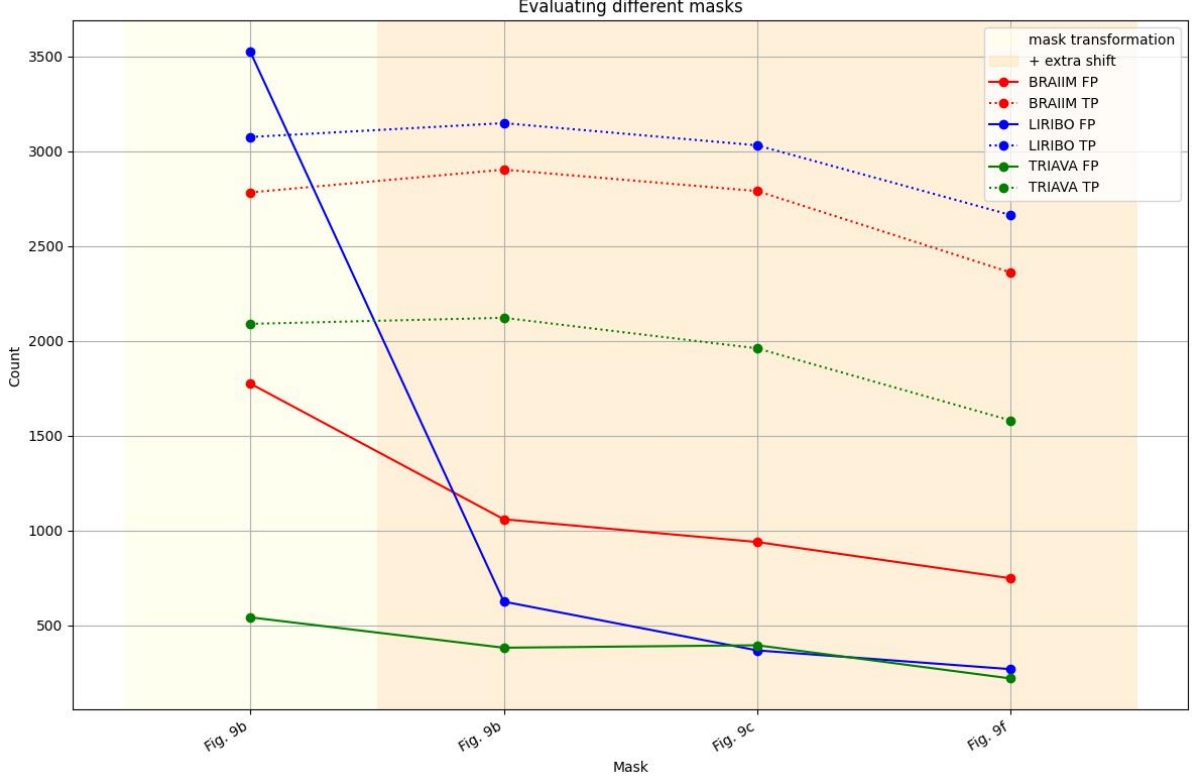


Figure 14: Evaluating the masks given in Figure 8. The thicker lines of mask 8c in combination with the transformation (i.e. adapting the mask to each single image) reduces false positives by a large margin. Even thicker lines tend to also decrease the true positives. The extra shift of the mask mainly reduces the number of false positives for the leaf miner flies.

Training Set	Metric	Fungus Gnats	Leaf Miner Flies	Whiteflies
Labeled Training Set	95th percentile box size	42260	23544	3275
	95th percentile box ratio	1.74	1.77	1.88
Unlabeled Training Set	95th percentile box size	43456	28480	3477
	95th percentile box ratio	1.73	1.76	1.84

Table 10: Found possible thresholds for labeled and unlabeled training set.

The mask with the best results (Figure 8d) produced distributions of boxes sizes, deviations from 1:1 ratio of sidelenghts and distribution of values for the whitefly rectangles, all taken from the results of the simple filter set 1 (see Table 5. Table 11 gives the found values for the metrics for both the labeled and the unlabeled training set, which are similar as expected.

3.4.2 Segmentation

Applying a refined filter set with thresholds derived by statistics reduced the number of false positives further and increased precision while retaining a sufficient number of true positives for the following self-training.

Figure 15 shows the results of applying the filters. Table 11 shows the resulting numbers of rectangles created by the image processing.

Figure 15 shows the results for the tests with the found thresholds. The two filter sets 1 and 2 as given in Table 5 are always the first and last steps in the graphs. In between single changes to set 1 are visualized. For the fungus gnats, all tried changes yield smaller effects by tolerable reductions of the number of true positives and were combined in the end. The number of false positives is still the highest compared to the other classes, which as already observed in the evaluation of the supervised training might be mainly due to missing labels. For the leaf miner flies, the effect are comparable but less prominent as in the fungus gnats, but precision and number of false positives were also already acceptable only with the first filter set. For the whiteflies, filtering the rectangles by box size and ratio was not taken over into the filter set, because it reduced the number of already scarce true positives too much. Instead overlapping boxes were deleted and filtering by value of the rectangles reduced false positives and at the same time retained the number of true positives.

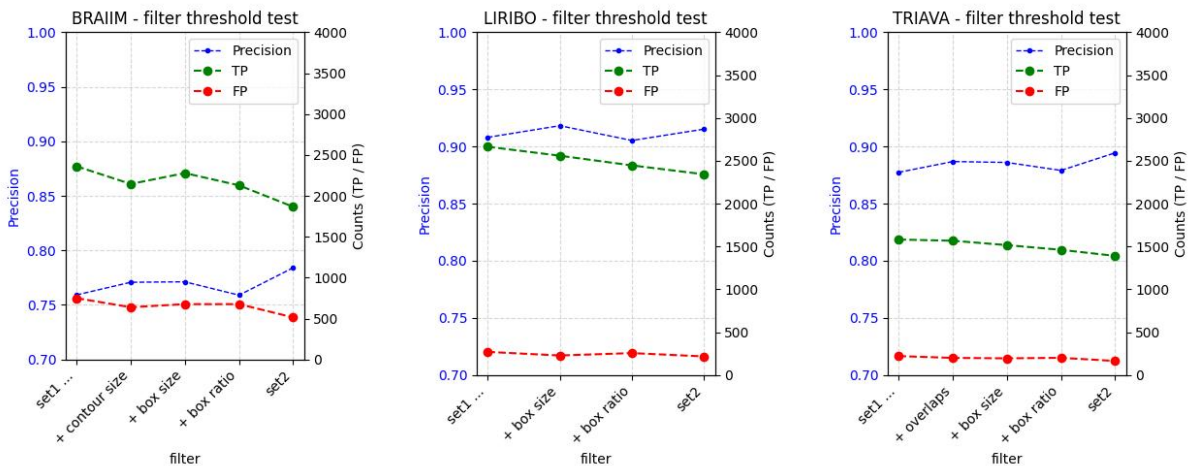


Figure 15: Results for different filters for each species. Leftmost is always the parameter set 1 with no extra filtering and estimated thresholds, rightmost is parameter set 2 with derived threshold and all extra filters applied. In between single additions to set 1 are tested. **Species do not follow my usual naming convention**

As to be expected objects of the same color, shape and size cannot be filtered out. This

concerns mainly parts of the insects as well as malformed or very untypical instances, as well as insects occluding each other. Table 11 gives the absolute numbers of rectangles returned which each filter set for the labeled and unlabeled training set. As expected they turn out to be very similar with exception of the total numbers of whiteflies, which is doubled in the unlabeled training set. As the number of filtered out rectangles is also proportionally larger, this is possibly only a effect of more crowded whitefly images in the unlabeled set.

Class	Labeled training set		Unlabeled training set	
	Filter set 1	Filter set 2	Filter set 1	Filter set 2
Fungus gnats	3110	2380	3477	2699
Leaf miner flies	2934	2684	3116	2628
Whiteflies	1802	1575	5015	4308

Table 11: Absolute number of rectangles created.

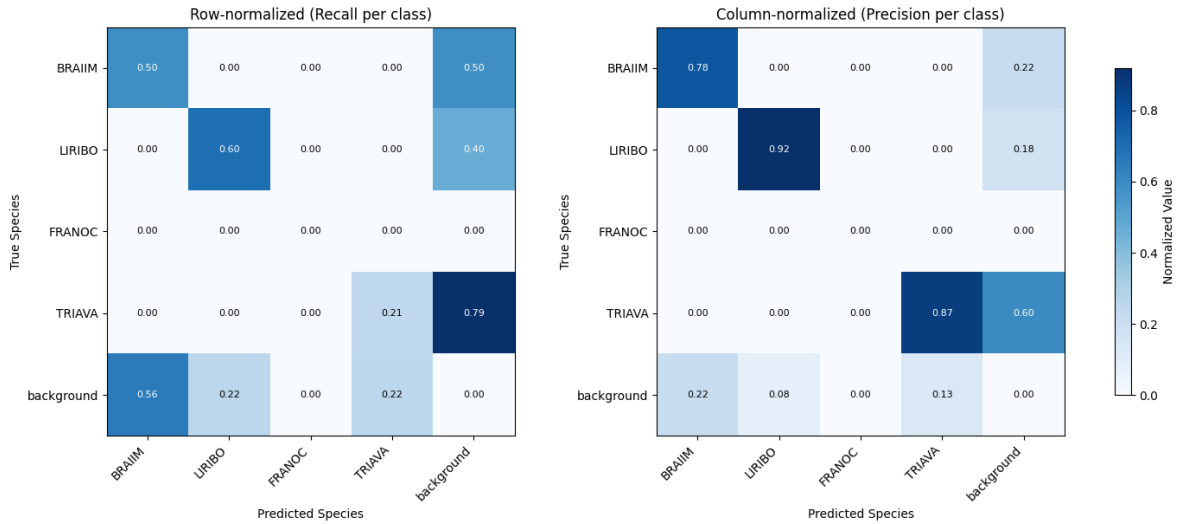


Figure 16: Confidence matrices for the image processed masks. Low recall is mainly due to the insects on the background structures that were masked out.

3.5 Self training

A model trained on the augmented data sets 1 & 2 as full images learned the class distinctions and also predicts more individuals, especially for objects on background structures (compare Fig 12 in the final results with the Fig 16 above). The model is used to predict pseudo labels and is then trained on the original images, which improves its performance only slightly and only for one epoch.

- The best model trained on the labels created by image processing used the following

parameters: 20 epochs, parameterset 2 with additional flipud augmentation set top 0.5, and both augmentation set 1 and 2 combined - same setting for full images and tiles (which included some empty tiles)

- The best model was used to predict on the original images. Predictions were used as pseudo labels with confidence thresholds taken from visual inspection of the predictions. Then the model was trained for another 20 epochs, but now on the original data. This was repeated two times (for full images) and three times (for tiles).

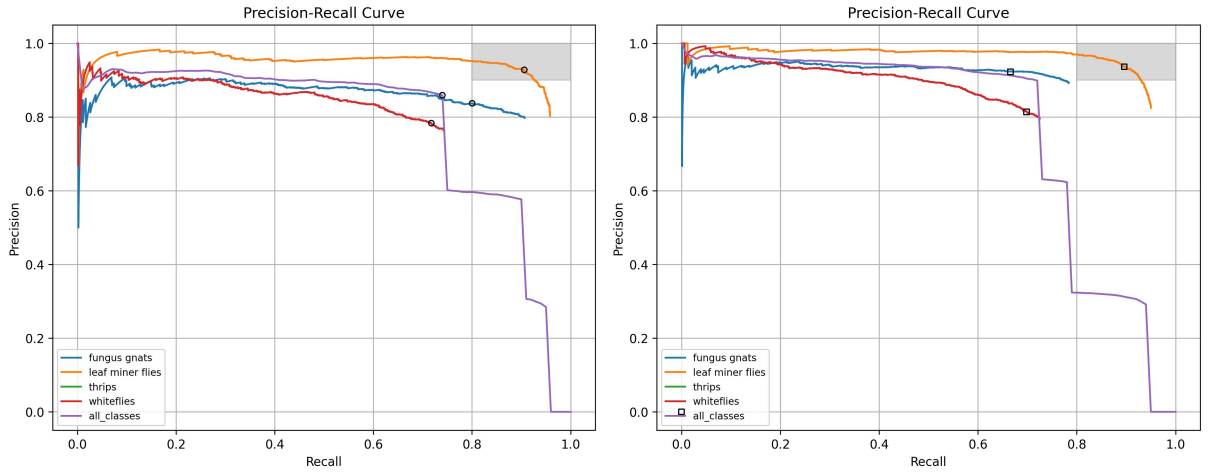


Figure 17: Precision-recall curves for the best model self-trained on tiles, the grey rectangle defines the target zone. The left curve is taken on the validation set with the operating points closest to the target zone marked as circles. The right curve is taken on the test set with the operating points taken from the validation set marked as squares. The results are worse then training on full images (compare Fig 11)

4 Discussion

- for single application, the brute force method is the most efficient solution given coherent and complete labeling for at least the validation and test set.
- for tables without background and with clean objects and good contrast, the image processing/self training approach will possibly work much better
- Quality of the data is key: no background, clean images, correct labeling
- Other problems still to be solved: Final images have lower res (domain shift), individuals sinking into the glue and changing their appearance
- grounding models might be a viable alternative

- not yet tried: tile size 1280. doing fully automated self training.
- creating pseudo-labels from the labels created by image processing works seamless - after one training one them the models detects can distinct between classes and detects also the insects on background structures. However confidence thresholds do not allow to distinguish between true and false positives, e.g. foliage is detected as insect with high confidence. This prevents the model to predict good pseudo-labels. As the leaf miner set shows, clean images are key for good results.

5 Conclusion

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